Open University Learning Analytics: Improving Student Success & Retention

Introduction

The Open University is one of the world's largest distance-learning institutions, serving a diverse student population across age groups, socioeconomic backgrounds, employment statuses, and study intensities. As online education continues to expand, the institution's strategic focus has evolved from monitoring simple activity metrics—such as login frequency or click counts—to evaluating Academic Success Quality.

The core challenge is no longer measuring how much students interact with the Virtual Learning Environment (VLE), but rather how effectively those interactions translate into meaningful academic outcomes and sustained retention. High activity does not always equate to high performance, and excessive effort without corresponding results may indicate inefficiencies in learning behavior or structural challenges within modules.

This project analyzes The Open University's Learning Analytics dataset—including student demographics, registration data, course structures, assessment results, and VLE interaction logs—to construct and evaluate a new institutional performance metric: Academic Success Efficiency (ASE).

By examining the relationship between learning effort and academic performance, this analysis aims to uncover behavioral, demographic, and structural drivers of efficiency, identify at-risk learners early, and provide actionable insights to improve retention, academic outcomes, and long-term institutional performance.

Problem Statement

The Open University's performance has traditionally been assessed using engagement-based metrics such as total VLE clicks, login frequency, and assessment submissions. However, these activity-based indicators fail to capture whether student effort translates into strong academic results.

Low academic efficiency leads to:

- High withdrawal rates
- Assessment failures
- Repeated module attempts
- Reduced student retention
- Lower long-term institutional performance

The institution requires a structured, data-driven framework to evaluate **how efficiently students convert learning effort into academic success**.

This project introduces the metric:

Academic Success Efficiency (ASE)

ASE=Academic Performance (AP)÷Learning Effort (LE)

The objective is to:

- Identify student segments with **high ASE** (efficient learners)
- Detect **low ASE + high effort** patterns (struggling students)
- Recognize **low effort + low ASE** patterns (disengaged or at-risk students)
- Understand how demographics, study behaviors, and course structures influence efficiency
- Surface early warning indicators of withdrawal risk

By analyzing the Learning Analytics dataset, this project seeks to move beyond volume-based engagement metrics toward outcome-driven efficiency analysis that supports academic quality and retention improvement.

Analysis Approach

• **Understand the Business Objective:**
Define Academic Success Efficiency (ASE) as the primary performance metric and shift focus from activity volume to outcome efficiency. Review dataset structure to understand relationships between student demographics, enrollment patterns, assessment outcomes, and VLE behavior.

• **Define Metrics and Hypotheses:**
Construct Academic Performance (AP) using assessment scores, pass/fail status, final

results, and number of passed assessments.

Construct Learning Effort (LE) using total VLE clicks, active study days, assessment submission count, registration duration, and module load.

Develop testable hypotheses such as:

- “Higher early engagement leads to higher ASE.”
- “High effort without structured preparation leads to lower efficiency.”
- “Students with higher education background exhibit higher ASE.”
- “Late registration correlates with withdrawal risk.”

• **Data Cleaning and Preparation:**

Merge datasets using common keys (id_student, code_module, code_presentation, id_assessment).

Handle missing values in scores and registration data.

Remove duplicates and validate assessment attempts.

Convert categorical variables appropriately and standardize data types.

Detect and manage outliers in click counts and scores using IQR or percentile capping.

• **Feature Engineering and Normalization:**

Normalize performance and effort components using min-max scaling or z-score standardization.

Create derived features such as:

- VLE Intensity Index (Clicks ÷ Active Days)
- Early Engagement Score (Clicks in first 30 days)
- Assessment Preparation Ratio (Pre-assessment clicks ÷ Total clicks)

  Construct composite AP and LE indices before calculating ASE.

• **Exploratory Data Analysis (EDA):**

Analyze distributions of ASE, AP, and LE.

Segment ASE by demographics (age_band, gender, region, highest_education), socioeconomic factors (disability, IMD band), and course structure (module, presentation, credits).

Visualize effort vs performance relationships using scatter plots and density distributions.

Examine module-level difficulty variance and assessment type performance (TMA, CMA, Exam).

• **Hypothesis Testing:**

Test effort–performance relationships using correlation analysis and regression modeling.

Compare ASE across demographic and socioeconomic groups using ANOVA or t-tests.
Evaluate early engagement impact on final outcomes through grouped comparisons.

• **Retention Risk Analysis:**
Identify patterns such as:

- Low early engagement + low submission rate
- High effort but declining scores
- Late registration + low VLE usage
  Analyze withdrawal rates by module and demographic segments.

• **Correlation and Driver Analysis:**
Use Spearman correlation for numeric relationships.
Apply eta-squared or ANOVA for categorical drivers of ASE.
Prioritize features that most strongly influence efficiency and withdrawal risk.

• **Summarize Actionable Insights:**
Translate findings into recommendations to:

- Improve early engagement strategies
- Optimize assessment structures
- Provide targeted academic support
- Reduce withdrawal rates
- Improve Academic Success Efficiency across modules

This structured approach ensures the analysis directly supports institutional decision-making, enabling The Open University to enhance academic quality, optimize learning design, and strengthen long-term retention.

Dataset:

The dataset includes multiple tables covering essential metrics across academic, behavioral, and socioeconomic dimensions.

Key Variable Categories

1. Student Demographics & Socioeconomic Attributes (*studentInfo*)

This table captures student-level background characteristics, enabling segmentation analysis across demographic and socioeconomic groups.

Key variables include:

- id_student (unique student identifier),gender,age_band,region,highest_education, imd_band (Index of Multiple Deprivation band),disability,studied_credits,final_result (Pass, Distinction, Fail, Withdrawn),num_of_prev_attempts.

2. Registration & Enrollment Patterns (*studentRegistration*)

This table provides insights into student enrollment timing and duration of study.

Key variables include:

- id_student,code_module,code_presentation,date_registration,date_unregistration

These fields allow calculation of:

- Registration duration,Late registration patterns,Early withdrawals.

3.Course Metadata (*courses*)

This table defines the structural characteristics of each module.

Key variables include:

- code_module,code_presentation,module_presentation_length

This enables analysis of:

- Module-level duration differences,Presentation-level comparisons,Structural influence on ASE

4.Assessment Structure (*assessments*)

This table describes assessment design and weighting within each module.

Key variables include:

- id_assessment,code_module,code_presentation,assessment_type (TMA, CMA, Exam),date,weight

This dataset supports:

- Weighted performance scoring,Assessment-type comparison,Module difficulty evaluation

5. Student Academic Performance (*studentAssessment*)

This table records student performance in individual assessments.

Key variables include:

- id_assessment,id_student,date_submitted,is_banked,score

This dataset forms the foundation of **Academic Performance (AP)** by enabling:

- Average score calculation,Pass/Fail classification,Attempt analysis,Score progression tracking

6. VLE Activity Logs (*studentVle*)

This table captures granular student interaction with online learning materials.

Key variables include:

- id_student,code_module,code_presentation,id_site,date,sum_click

This dataset supports measurement of **Learning Effort (LE)** through:

- Total VLE clicks,Active study days,Temporal engagement trends,Pre-assessment activity patterns.

7. VLE Content Metadata (*vle*)

This table describes learning content available within the virtual environment.

Key variables include:

- id_site,code_module,code_presentation,activity_type,week_from,week_to

This enables:Content-type engagement analysis, Coverage rate calculations, Study behavior clustering by activity type

Research Hypotheses

H1: Effort, performance and withdrawal
Students with higher learning effort (LE_raw, clicks, active days) show higher academic performance (AP_raw), but this relationship differs for withdrawn and non-withdrawn students, so more effort is not always associated with higher ASE.

H2: Early engagement and efficiency
Students who engage more with the VLE in the first 30 days (early_clicks) achieve higher AP_raw and ASE_score than students with low early engagement.

H3: VLE intensity and efficiency
Students with moderate VLE Intensity (total_clicks ÷ active_days) exhibit higher ASE_score than students with very low or very high intensity, indicating an optimal range of daily online activity.

H4: Assessment preparation behaviour
Students with a balanced Assessment Preparation Ratio (moderate share of clicks before assessments) achieve higher AP_raw and ASE_score than students who prepare very little or concentrate almost all clicks before assessments.

H5: Assessment attempts and efficiency
Students who require repeated assessment attempts (high submission_rate_student or many resubmissions) have lower ASE_score than students who usually pass on the first attempt.

H6: Educational background and efficiency
Students with higher prior education (e.g. A-Level, HE or postgraduate qualifications) show higher ASE_score than students with lower or no formal qualifications.

H7: Socioeconomic background (IMD) and efficiency
Students from less deprived IMD bands (higher IMD percentiles) have higher ASE_score than students from more deprived bands, indicating a modest socioeconomic gradient in academic efficiency.

H8: Credit load and cognitive overload
Students enrolled in higher studied_credits bands have lower ASE_score and slightly lower AP_raw per module than students with lighter credit loads, consistent with cognitive overload when studying many credits at once.

Data Cleaning and Preparation

Removed exact duplicate rows from all base tables. Handled missing data by imputing sparse assessment dates with global medians, dropping highly missing VLE timing columns (week_from, week_to), treating missing IMD band as “unknown”, and merging all tables on (code_module, code_presentation, id_student) to create model_with_ASE

Outlier Analysis and Treatment

Inspected distributions of total_clicks and mean assessment scores using boxplots. Capped total_clicks at the 99th percentile and clipped mean assessment scores between lower and upper percentiles to reduce the impact of rare extremes while preserving typical variation.

Correlation Strength Analysis

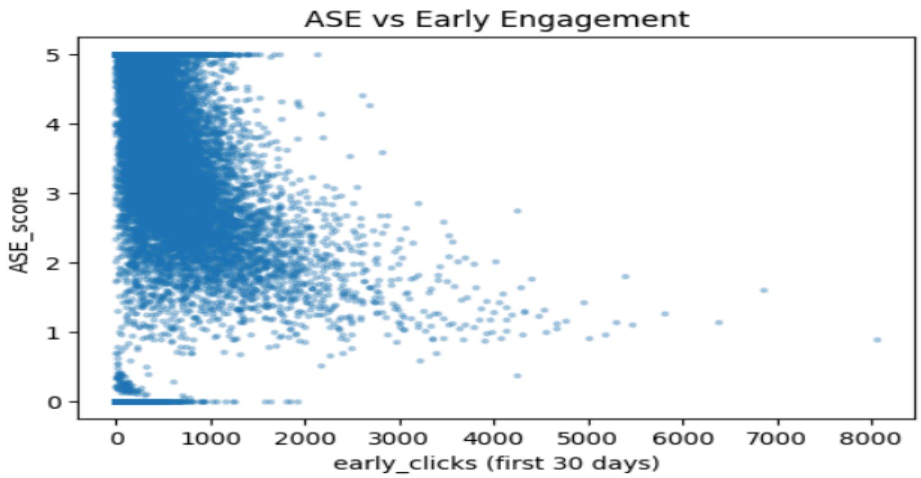
Computed correlation matrices for non-withdrawn students, showing AP_raw moderately correlated with LE_raw and total_clicks, while ASE_score correlated positively with AP_raw but weakly or negatively with LE_raw, clicks and active_days, indicating that high activity could signal inefficiency. For withdrawn students, ASE_score was more strongly aligned with both performance and effort, although even the relatively efficient within this group still withdrew, emphasizing that efficiency alone did not guarantee persistence.

Exploratory Data Analysis: Key Findings

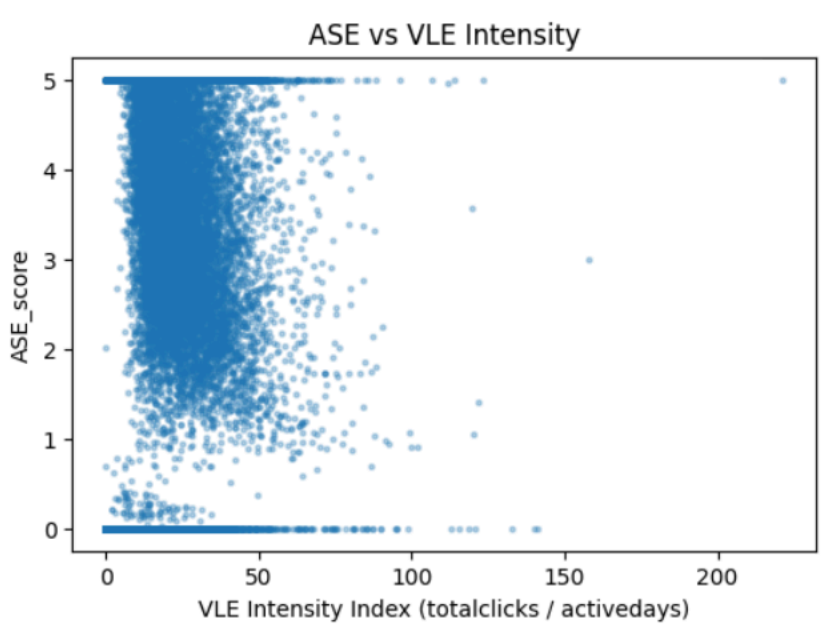
H1: Effort, performance and withdrawal – Partially Supported

Higher effort and VLE engagement were associated with higher performance in both withdrawn and non-withdrawn groups, but ASE did not always increase with effort: for non-withdrawn students ASE tended to decrease at very high levels of effort and activity, and withdrawn students with higher effort still left the course.

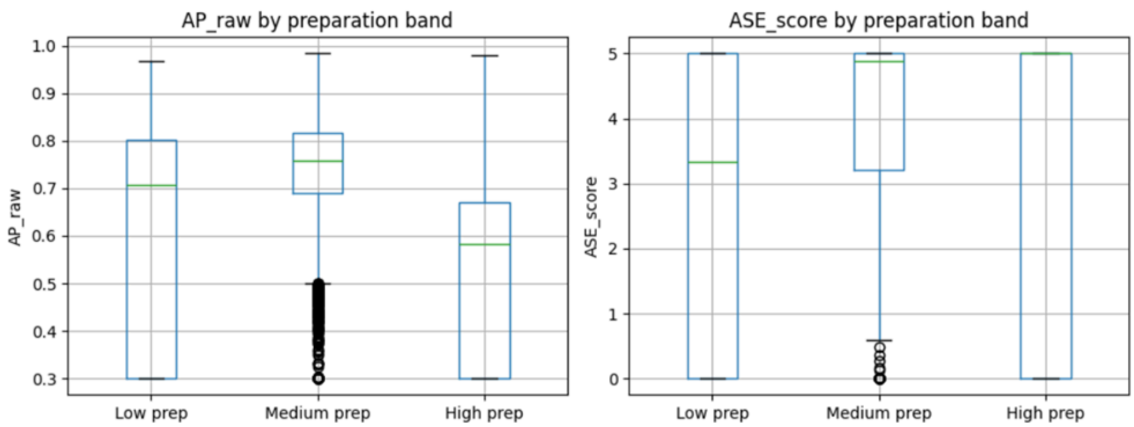
H2: Early engagement and efficiency – Partially Supported Students with stronger early engagement (higher early_clicks in the first 30 days) generally achieved higher AP and mid-to-high ASE compared with very low-engagement students, although efficiency still varied widely within each early-engagement band.



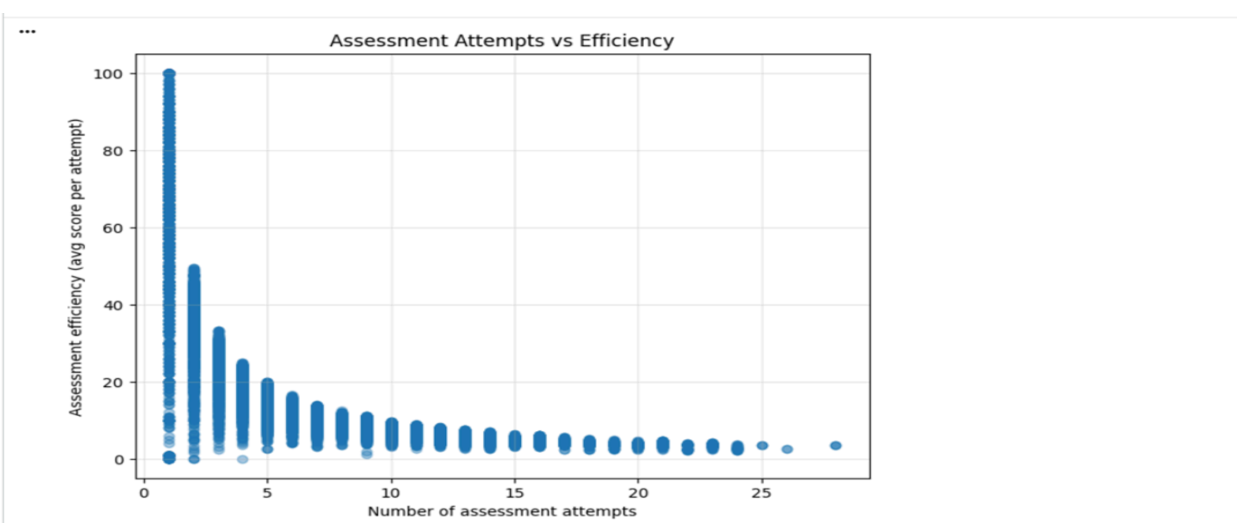
H3: VLE intensity and efficiency – Partially Supported Across the full range of VLE Intensity, ASE spanned from 0 to 5, but many efficient students clustered around low-to-moderate intensity, while very high-intensity students often showed only moderate ASE, indicating that extreme intensity was frequently inefficient.



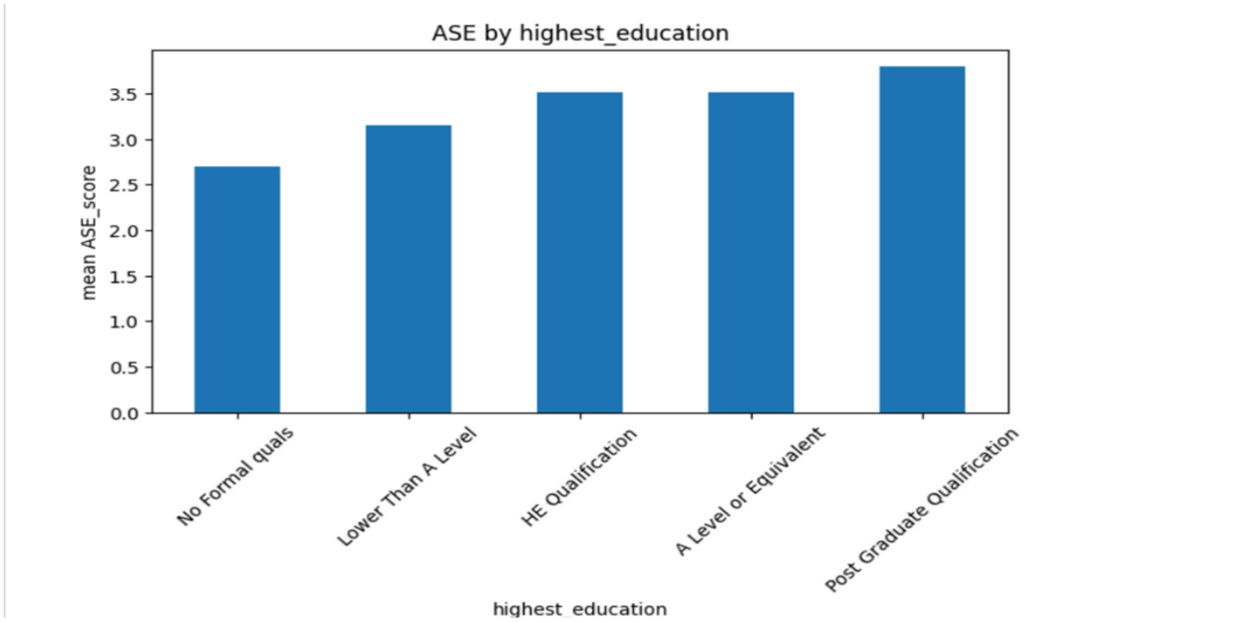
H4: Assessment preparation behaviour – Partially Supported When we grouped students by assessment preparation ratio, the medium-prep band showed the highest AP_raw and ASE_score, whereas low-prep students performed and behaved less efficiently, and high-prep students did not improve performance and had lower ASE than the medium group.



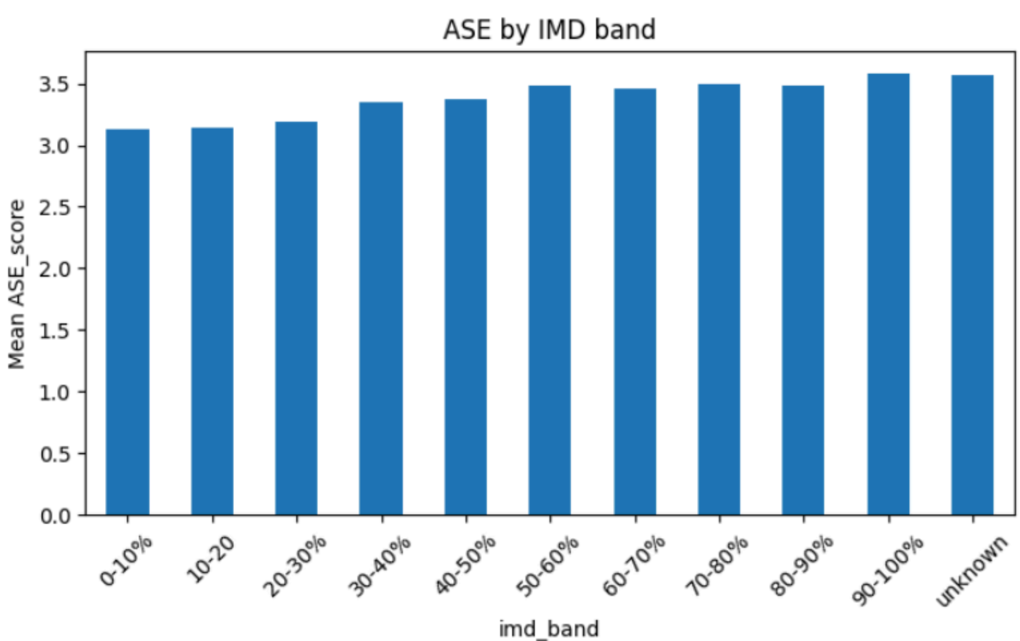
H5: Assessment attempts and efficiency – Supported Students with higher submission rates and more repeated assessment attempts typically had lower ASE_score than those who passed on the first attempt, indicating that extra retries improved marks only at the cost of disproportionately more effort.



H6: Educational background and efficiency – Supported ASE_score increased stepwise from students with no formal qualifications to those with postgraduate qualifications, showing that stronger prior education was linked to more efficient conversion of effort into performance.



H7: Socioeconomic background (IMD) and efficiency – Supported ASE_score rose gradually from the most deprived IMD bands to the least deprived bands, so students from less deprived areas were somewhat more efficient than those from highly deprived areas, even though the differences were moderate.



H8: Credit load and cognitive overload – Supported Students in the low-credit band achieved the highest ASE_score and AP_raw, while both metrics declined in the medium- and high-credit bands; LE_raw per module was also lowest for high-credit students, consistent with effort being spread too thinly across many modules.

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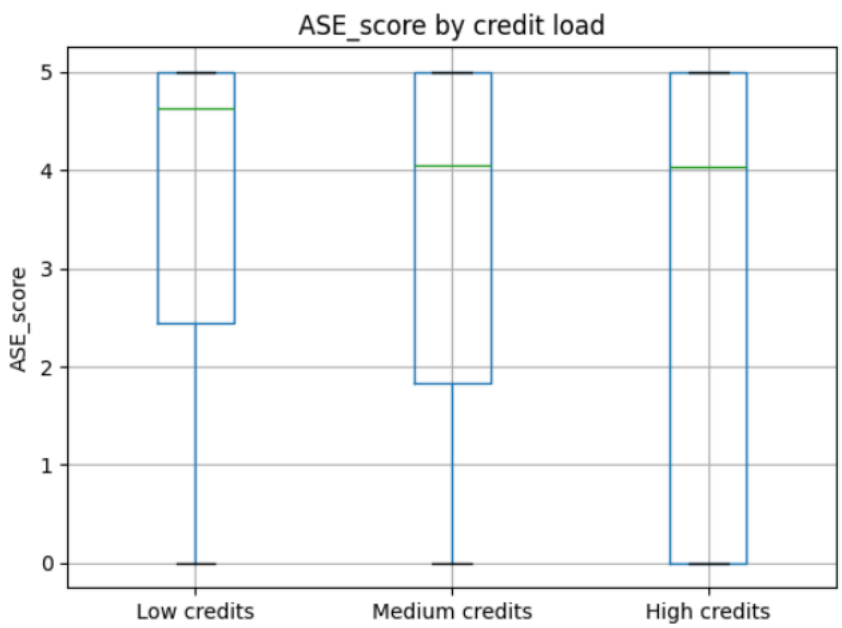

Summary and Recommendations

Analysis indicated that more effort generally improved performance but not always efficiency, especially for non-withdrawn heavy users whose extra clicks yielded diminishing returns in ASE. Balanced behaviours (moderate early engagement, moderate VLE intensity, medium assessment preparation ratio) were consistently associated with the highest ASE, while students with weaker prior education, higher deprivation or heavy credit loads showed lower efficiency and appeared to need additional support.

Actionable Recommendations

- Track ASE alongside AP and LE in learning analytics dashboards to flag students who are working hard but inefficiently and provide targeted study-skills and feedback interventions.
- Encourage balanced engagement through pacing guidance, formative checkpoints and communications that discourage both last-minute surges and unproductive over-clicking.

- Use registration timing, early engagement and repeated assessment attempts as early-warning indicators to trigger proactive outreach to late registrants, low-prep students and heavy re-submitters.
- Provide additional academic and pastoral support for students with low prior education, from more deprived IMD bands, or carrying heavy credit loads, for example via preparatory modules, workload planning advice and targeted tutoring.